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Research article

Green roof storage capacity can be more important than evapotranspiration for retention performance

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A B S T R A C T

Green roofs can significantly reduce stormwater runoff volumes. Plant selection is crucial to retention performance, as it is influenced by how well plants dry out substrates between rainfall events. While the role of plants in evapotranspiration (*ET*) on green roofs is well-studied, their potential influence on retention via their impacts on water movement through substrates is poorly understood. We used a simulated rainfall experiment with plant species with different water use strategies to determine the key drivers of green roof retention performance. Overall per-event retention was very high (89–95%) and similar for all plant species and unplanted modules for small events. However, for larger events, some species showed lower retention than unplanted modules or low-water using succulent species. Despite the fact that these species were more effective at replenishing storage between rainfall events due to their higher *ET*, they reduced the maximum storage capacity of the substrate, likely due to their root systems creating preferential flow paths. This finding has important implications for green roofs, as although *ET* represents the primary means by which the storage capacity of green roofs can be regenerated, if species with high *ET* also reduce the maximum storage capacity, effective retention performance is reduced. Therefore, we suggest that species selection must first focus on how plants affect storage capacity in the first instance and consider water use strategies as a secondary objective.

1. Introduction

The creation of impervious areas as part of urbanisation increases surface runoff and disturbs the natural flow regime of streams (Poff et al., 1997), resulting in the degradation of receiving ecosystems (Madsen et al., 2009; Walsh et al., 2005) and causing flooding. To return towards a more natural flow regime, surface runoff should be reduced by ~60–80%, equivalent to pre-development losses via evapotranspiration (*ET*) (Burns et al., 2012; Zhang et al., 2001). Green roofs are among the more promising technologies employed to reduce surface runoff, as roofs constitute a large proportion of impervious area in dense urban areas and do not compete with ground-level development for space (Arnold Jr and Gibbons, 1996). Green roofs can help compensate for the loss of natural landscapes and hydrological processes by intercepting and retaining rainfall which is used to sustain plant growth, restore *ET* and therefore reduce runoff (Stovin et al., 2012; Viola et al., 2017).

Green roofs have been shown worldwide to be an effective stormwater management tool, although their performance varies. Previous studies have shown annual rainfall retention from 5 to 90% (Carter and Rasmussen, 2006; Cipolla et al., 2016; Locatelli et al., 2014; Sims et al., 2016; Szota et al., 2017b). Per-event retention varies from 5 to 100% (Bliss et al., 2009; Carson et al., 2013; Locatelli et al., 2014; Sims et al., 2016; Stovin et al., 2015). Climatic conditions and differences in green roof configuration, i.e., substrate type and depth, as well as plant species selection, drive this variation in performance (Berndtsson, 2010). Rainfall characteristics (size, intensity and temporal distribution) and reference evapotranspiration (ET_o) dynamics are particularly important drivers of green roof rainfall retention and runoff across Mediterranean (Fioretti et al., 2010; Viola et al., 2017), temperate (Fassman-Beck et al., 2015; Stovin et al., 2012) and tropical climates (Voyde et al., 2010). Climates with a high frequency of small rainfall events combined with high evaporative demand tend to have higher rainfall retention (Carpenter and Kaluvakolanu, 2011; Elliott et al., 2016; Sims et al., 2016; Speak et al., 2013). The variation in retention performance has also been explained by green roof substrate water holding capacity, which is dependent upon substrate characteristics such as the proportion of different materials and depth (De-Ville et al., 2017; Farrell et al., 2013a; Hilten et al., 2008; Morgan et al., 2013; VanWoert et al., 2005; Voyde et al., 2010). *ET* is influenced by substrate characteristics, depth and species selection (Farrell et al., 2013b; Nagase and Dunnett, 2012; Szota et al., 2017a; Wolf and Lundholm, 2008). Therefore, on an event

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basis, retention will be determined by substrate water content before the rainfall event, the size of the rainfall event and the amount of water available for *ET* after the event.

As such, retention performance will ultimately depend on the interactions between climatic conditions (rainfall and ET_o), green roof configuration (substrate depth and properties, plant species, protection and drainage layers) and available storage conditions before rainfall events (Sims et al., 2016; Stovin et al., 2015; Viola et al., 2017). However, few studies have investigated the relative importance of these different parameters on retention and how they interact with each other. Locatelli et al. (2014) identified that the most sensitive parameters for long-term retention performance were crop coefficients (ratio between actual *ET* and ET_o which depends on plant type) and sub-surface storage (water retention in the drainage layer). For single events, the most sensitive parameters were sub-surface storage and initial substrate moisture conditions. However, Locatelli et al. (2014) primarily investigated *Sedum* species, which are low water-using species (Starry et al., 2014), and it is possible that other species with different water-use strategies might change which parameters are the most important for event-based retention.

Several studies have explored the impact of alternative species to *Sedum* on retention (Buccola and Spolek, 2011; Graceson et al., 2013; Szota et al., 2017b), but their results were not conclusive or consistent. Succulent species such as *Sedum* outperform non-succulent species during drought due to their conservative water use. Although they have relatively high biomass, succulents show low canopy interception and *ET* and generally show lower rainfall retention (Azeñas et al., 2018; Li et al., 2018). In contrast, non-succulent species show significant potential for runoff mitigation (Soulis et al., 2017a). Considering that plants can significantly influence green roof retention performance via *ET* (Li et al., 2018), several authors have suggested that selecting plants with high *ET* should improve retention performance (MacIvor et al., 2011; Poë et al., 2015; Szota et al., 2017a; Whittinghill et al., 2015).

However, the influence of plants on green roof retention is potentially due to factors other than *ET* (Fassman-Beck et al., 2013; Morgan et al., 2013; Ouldboukhitine et al., 2012). These additional factors include leaf succulence, which adds extra usable water (Farrell et al., 2012; Wadzuk et al., 2013), and plant roots, which can fill large substrate pores to increase substrate water capacity (Nagase and Dunnett, 2011; Poë et al., 2015). In our previous study (Zhang et al., 2018), we found that the root systems of some plant species reduced substrate water holding capacity by introducing preferential flow paths. Therefore, while previous studies have started to pay attention to non-*ET* plant effects on retention, there are only few studies looking at how these factors might interact to influence retention performance. Consequently, it may mean that the importance placed on selecting plants solely with high *ET* to achieve higher rainfall retention may lead to inappropriate plant selection for stormwater mitigation on green roofs.

To assist with green roof plant selection, we therefore need to better understand the relative importance of plant-related factors on green roof hydrological processes and therefore retention performance. In this study, we determined to what extent non-*ET* plant effects, i.e., the reduction in substrate storage capacity by plant roots, affected retention relative to how efficiently plants replenished storage capacity in the substrate via *ET*.

2. Materials and methods

2.1. Experimental design

This study used a simulated rainfall setup with green roof modules (Fig. S2), where components of the water balance could be measured, including runoff after each rainfall event, substrate moisture before and after rainfall events and *ET* between rainfall events. This experiment was set up at the Burnley Campus of The University of Melbourne (−37.828472, 145.020883). The experiment ran for 468 days, from 14 July 2015 to 25 October 2016.

Eighteen green roof modules (1.15 × 1.15 m) were arranged in a randomised block design under a raised, open-ended canopy (28 m long, 5 m wide and 2.5 m tall), that excluded rainfall without raising temperatures appreciably. To simulate rainfall events, it was fitted with an overhead irrigation system that used pressure-regulated drippers on a 10 cm grid, positioned 1.32 m above the surface of each module. The irrigation system applied simulated rainfall at a consistent intensity of 45.7 mm h^{-1} to each module. This rainfall intensity is very high and similar to the 1% annual exceedance probability for Melbourne (48.6 mm h^{-1} for a 1 h duration; Australian Bureau of Meteorology, 2018). Each green roof module consisted of a plastic base (solid Nally MegaBins®; 780 L capacity, 1165 × 1165 × 780 mm) which supported a steel frame which held a plastic tray containing the green roof layers, a drainage layer (VersiDrain® 25P; water storage capacity 6.1 L m^{-2}; equivalent to 8.1 mm), a geotextile fabric layer (Elmich A14) and a scoria-based substrate layer with a depth of 100 mm. Before planting, laboratory tests of substrate properties were carried according the Australian Standard for Potting Mixes AS 3743-2003 (Standards Australia, 2003), including air-filled porosity (13.8%), bulk density (1.26 g cm^{-3}) and water holding capacity (45.9%) (Farrell et al., 2013a). These properties comply with requirements of FLL guidelines for green roof substrates (FLL, 2008). Substrate properties were re-measured at the end of the experiment and there was no significant difference in the water-holding capacity of the substrate between the start and end of the experiment (Zhang et al., 2018).

Each module was randomly allocated to one of six ‘vegetation’ treatments and there were three replicates of each treatment: (1) bare un-planted substrate (as a control); (2) *Sedum pachyphyllum*; (3) *Lomandra longifolia*; (4) *Dianella admixta*; (5) *Stypandra glauca*; (6) mixture of *Lomandra longifolia, Dianella admixta* and *Stypandra glauca.* Eighteen tubestock-sized plants were planted in each vegetated module in June 2012. These plants have all been shown to be appropriate species for green roofs in southern Australia (Farrell et al., 2013b). Vegetation cover was maintained during the experiment, with little seasonal variation, and no plants were replaced during the study period.

2.2. Meteorological conditions during the experiment

Meteorological data for air temperature, solar radiation, wind speed and relative humidity were sourced from Olympic Park, Melbourne, Australia (Melbourne Regional Office; site 086071; Australian Bureau of Meteorology, 3 km from the site). Reference *ET* (ET_o) between events was calculated as the cumulative ET_o during the dry periods between events, which is the *ET* from a standardised vegetated surface using the Penman-Monteith equation (FAO56-PM) (Allen et al., 1998).

2.3. Rainfall simulation

The applied rainfall regime simulated a ‘typical’ rainfall year in Melbourne, Australia, based on analysis of long-term rainfall data (1965–2015; Melbourne Regional Office; site 086071; Australian Bureau of Meteorology). To create a ‘typical’ rainfall year and retain natural variability in event size distribution, the 50^{th} percentile rainfall depth for each season was identified and then the four seasons were combined to create a ‘typical’ year. The frequency of each simulated rainfall event size matched well with the long-term rainfall frequency of event sizes for Melbourne (Fig. 1a.). A total of 92 rainfall events were applied over the 15 months of the study and data was collected for 65 of these events. The 25, 50, 75, 90, 95 and 99th percentile daily rainfall depths were 0.6, 2.0, 5.0, 11.0, 16.8 and 35.2 mm (Fig. 1a and b). The distribution of antecedent dry weather periods (ADWP) in the simulated rainfall experiment was also similar to the long-term average (Fig. 1c) and the 25, 50, 75, 90, 95 and 99th percentile ADWPs were 0, 2, 4, 6, 9 and 12 days (Fig. 1c and d). Similarly, the distribution of ET_o

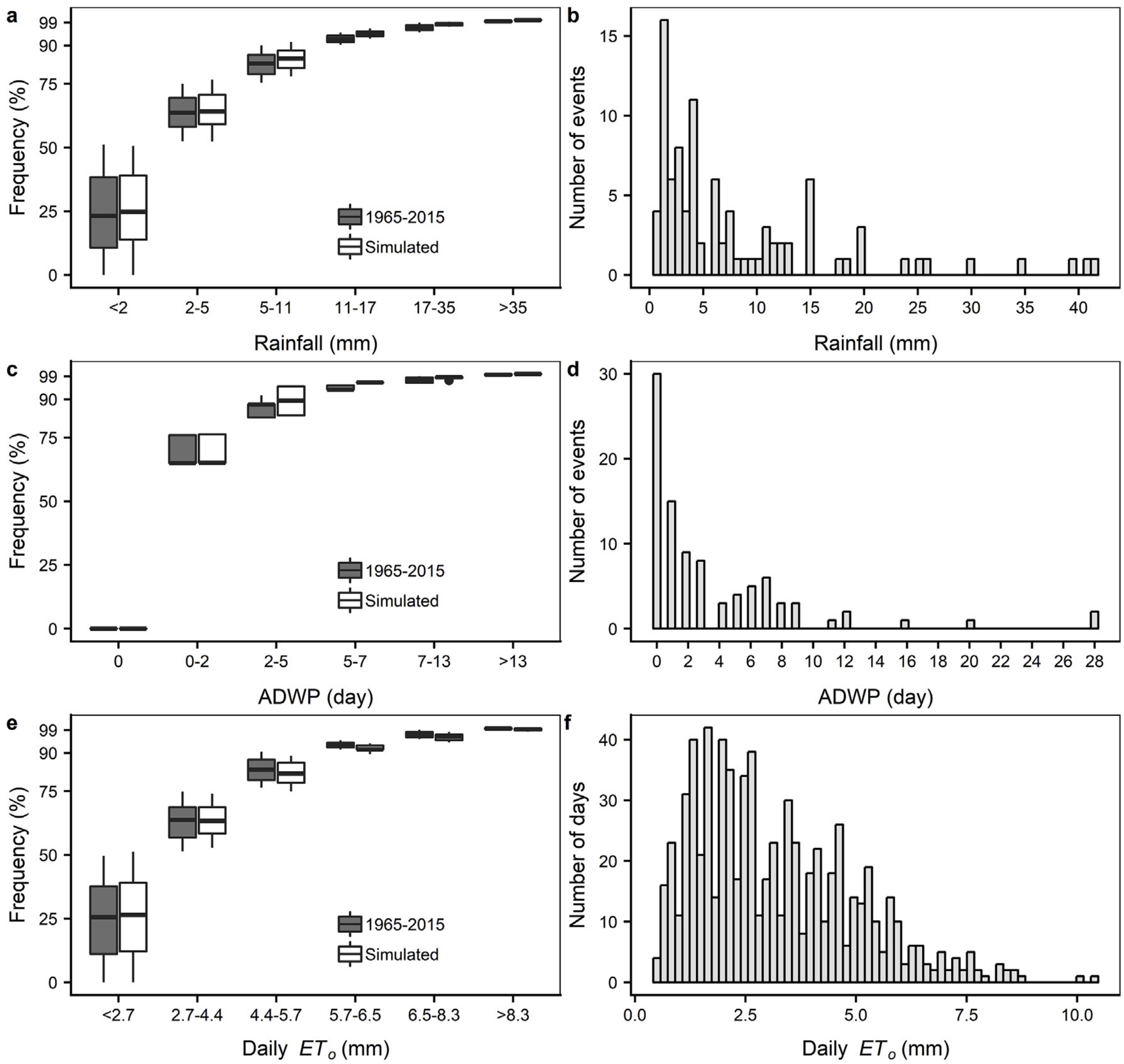

Fig. 1. Distributions of historical rainfall (1965–2015) and the simulated rainfall applied in this study (a and b); distributions of historical ADWP (1965–2015) and the simulated ADWP applied in this study (c and d) and distributions of historical daily ET_o (1965–2015) and daily ET_o during this study (e, f).

in the simulation were similar to the long-term average (Fig. 1e) and the 25, 50, 75, 90, 95 and 99th percentile ET_o values were 1.7, 2.7, 4.4, 5.7, 6.5 and 8.28 mm (Fig. 1e and f).

2.4. Runoff collection and calculating rainfall retention per event

Runoff from each module was collected in a U-shape PVC pipe for each simulated rainfall event. One side of the 'U' was connected to the module outlet and the other side was fitted with a 0.5 m Odyssey capacitance water level sensor (Dataflow Systems Ltd., Christchurch, NZ) to measure runoff depth. Runoff depth (mm) was calculated as the total change in water level depth multiplied by the cross-sectional area of the pipe (150 mm inner diameter). Rainfall retention was calculated as rainfall depth minus runoff depth for each event (all in mm).

2.5. Maximum module water storage

The maximum storage capacity of each module was determined gravimetrically, using a load cell with 0.05 kg resolution (HBM RSCC s-type; Noise & Vibration Measurement Systems Pty Ltd (NVMS), WA, Australia) attached to a gantry crane. Maximum storage capacity ($S_{\max}$; mm) was determined as:

$$S_{max} = \frac{M_{max} - M_{min}}{A_m} \tag{1}$$

where $M_{\max}$ (kg) is the weight of the module at field capacity, $M_{\min}$ (kg) is the weight of the module when it was the driest in the season, and A_m is the surface area of the module (m^2). *Both* $M_{\max}$ and $M_{\min}$ were determined seasonally, to account for changes in plant cover and therefore mass, either due to growth or senescence.

2.6. Available water storage before each rainfall event

Available storage before each rainfall event (*AS*; mm) represents the depth of space in the substrate which can retain rainfall and was calculated as:

$$AS = S_{max} - \frac{(M_{before} - M_{min})}{A_m} \tag{2}$$

where M_{before} (kg) is the module mass measured before each rainfall event.

2.7. Substrate water content after each rainfall event

Once drainage had finished, the module mass was measured to determine the gravimetric substrate water content after rainfall (*SWC*;

mm):

$$SWC = \frac{M_{after} - M_{min}}{A_m} \tag{3}$$

where M_{after} (kg) is the weight of the module, taken 24 h after each event to avoid generating runoff by lifting the modules to obtain their weight.

2.8. Evapotranspiration between rainfall events

ET was the cumulative water loss during the dry periods between rainfall events and determined from module mass:

$$ET = \frac{M_{after\ n} - M_{before\ n+1}}{A_m} \tag{4}$$

where ET is evapotranspiration after event n and before event $n+1$ (mm), $M_{after\ n}$ is the module mass 24 h after application of event n (kg), $M_{before\ n+1}$ is the module mass prior to application of event $n+1$ (kg).

2.9. Relationship between ET/ET_o and substrate water content

The ability of plants to deplete stored moisture in the substrate was determined from the relationship between ET and SWC. As ET is also driven by ET_o on any given day, we corrected ET by dividing it by ET_o (i.e., ET/ET_o). Further, because the number of days between rainfall events was not constant, it was important when developing the relationship between ET/ET_o and SWC to only utilise ET/ET_o data where the time between event n and event $n+1$ was one day. This is important as using periods greater than one day may under-estimate ET/ET_o for a given SWC, especially where SWC is depleted at unknown point in time during that period.

2.10. Data analysis

One-way analysis of variance (ANOVA) in combination with Tukey's post-hoc test were used to determine differences among treatments per event retention, the number of events generating runoff, available storage in the substrate before each event and the maximum module storage capacity. Regression analysis was used to relate rainfall depth and available storage in the substrate to retention for each event, as well as to relate ET/ET_o to substrate water content. Multiple linear regression analysis combined with hierarchical partitioning was used to identify which plant-based variables had the greatest influence on available storage in the substrate before each rainfall event (Hatt et al., 2004; Walsh and Mac Nally, 2013). We used the 'hier.part' package for the hierarchical partitioning to describe the strength of relationship between available storage and plant-based variables (Walsh and Mac Nally, 2013). All data shown are non-transformed data. We used R for all statistical analysis (R Core Team, 2017).

3. Results

3.1. Per-event retention performance

Under simulated Melbourne rainfall conditions, 89–95% of rainfall was retained on an event basis for the 65 rainfall events measured (Table 1). Modules planted with *L. longifolia* showed 5.5% greater retention than modules planted with *S. glauca* modules, which had the lowest retention. For 92 applied rainfall events, modules planted with *L. longifolia* generated runoff in 20% of applied rainfall events and *S. glauca* modules had 29% of events generating runoff.

3.2. Relationship between rainfall depth, runoff and retention

Runoff was significantly related to rainfall depth ($P < 0.001$) and plant species had significant effects on the relationship between rainfall

Table 1
Mean (± SE; n = 3) per-event retention (%) and the percentage of rainfall events generating runoff. *P*-values indicate significant differences among plant treatments (one-way ANOVA; Tukey's post-hoc test).

Treatment	Retention (%)	Events generating runoff (%)
Bare	92.62^{ab} (± 0.22)	21.03^{ab} (± 1.02)
S. pachyphyllum	91.06^{bc} (± 0.52)	25.13^{ab} (± 1.02)
L. longifolia	94.80^{a} (± 0.18)	20.00^{b} (± 0.89)
D. admixta	92.83^{ab} (± 0.37)	25.64^{ab} (± 1.38)
S. glauca	89.34^{c} (± 0.30)	29.23^{a} (± 0.00)
Mixture	90.31^{bc} (± 0.52)	28.72^{ab} (± 0.51)
P-value	< 0.001	< 0.05

Different letters denote significant differences among means (one-way ANOVA, n = 3; P < 0.001).

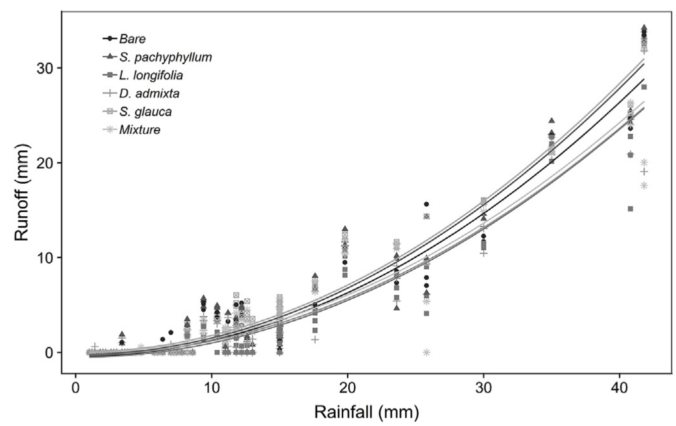

Fig. 2. Relationship between rainfall and runoff for all treatments. Symbol shape indicates plant treatments.

and retention ($P < 0.001$) (Fig. 2). The differences in retention among the plant species were significant in larger rainfall events and modules planted with *S. glauca* and the mixture, showed lower retention than *L. longifolia* modules (Fig. 3).

3.3. Relationship between available storage in the substrate and retention

Available storage in the substrate prior to each event was significantly related to retention, but only for events > 5 mm, which is intuitive (Fig. 4). For these larger events, available storage before the event was greatest for *L. longifolia* and lowest for *S. glauca*, which was consistent with their retention performance.

3.4. Maximum module storage capacity

The choice of plant species significantly influenced the maximum storage capacity of the modules (Fig. 5). Modules planted with *S. pachyphyllum* had the highest maximum water content, while modules with *S. glauca* and the mixture had 36% and 32% lower maximum water content than that of *S. pachyphyllum.*

3.5. Relationship between ET/ET_o and substrate water content

The ratio between ET and ET_o and substrate water content reflects the potential of each treatment to restore available storage via ET between rainfall events. Substrate water content, plant treatment and their interaction significantly influenced ET/ET_o ($P < 0.001$). The slope of the relationship between ET/ETo and substrate water content was significantly different between planting treatments ($P < 0.001$), which showed that species differed in their water sensitivity. These relationships showed that *S. pachyphyllum* was the most conservative water user, with its ET/ET_o having low sensitivity to water content

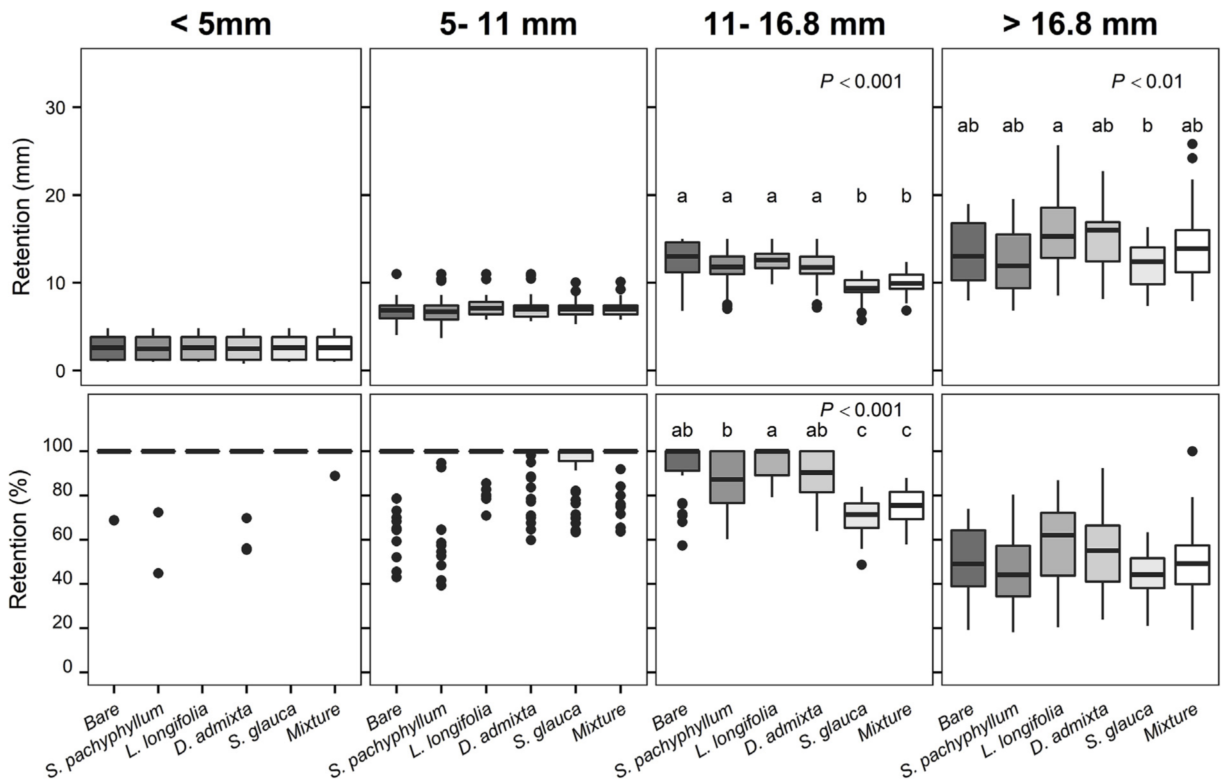

Fig. 3. Per-event retention depth (mm) and percentage (%) for plant treatments in relation to key rainfall depth percentiles. *P*-values indicate differences between treatments from one-way ANOVA.

(slope = 0.004) (Fig. 6). In contrast, *L. longifolia*, the species mixture and *S. glauca* all showed relatively higher sensitivity to substrate water content (slope = 0.022, 0.036 and 0.039, respectively) and therefore these species potentially had greater capacity to deplete substrate water at a higher rate between rainfall events than the conservative and insensitive species did. However, the maximum substrate water content was restricted to less than 15 mm for bare substrate, *S. glauca* and the mixture; whereas *D. admixta* and *L. longifolia* showed water content up to 20 mm (an increase of 33% compared to previous treatments). Substrate water content for *S. pachyphyllum* ranged between 10 and 25 mm (an increase of up to 66%).

3.6. Identifying the key plant-based driver of event-based retention performance

The frequency and depth of rainfall events are clearly the major drivers of rainfall retention. A multivariate model with rainfall depth, maximum storage capacity of each treatment and the evapotranspiration potential of each treatment revealed a high correlation of these parameters with rainfall retention on a per-event basis ($R = 0.84$; $P < 0.001$, data not shown). Of these variables, rainfall depth is dominant, with hierarchical partitioning showing that more than 68% of the variance in retention is explained by rainfall depth.

The significant effect of rainfall makes it difficult to discern the relative influence of plant-related explanatory variables. Further, while explaining the variance in retention performance is the ultimate aim of our study, including rainfall depth in the model does not elucidate the main plant-based driver of retention. Therefore, we created a multiple linear regression model to predict available storage in the substrate using only the two plant-based explanatory variables: i) the maximum storage capacity and ii) the evapotranspiration potential of each treatment. The available storage was explained 61% by maximum water content and ~39% by evapotranspiration potential (Fig. 7). Despite not having rainfall as an explanatory variable, this model produced fitted values which were strongly correlated with available storage ($R = 0.56$ and $P < 0.001$).

4. Discussion

4.1. Per-event retention performance

In this study, mean per-event retention ranged from 89 to 95%, which is relatively high compared with values reported in the literature (Locatelli et al., 2014; Loiola et al., 2018; Sims et al., 2016; Stovin et al., 2012), which most likely reflects the influence of different climates and green roof configurations (Berndtsson, 2010). Interestingly, although retention was high overall, the differences among plant treatments for the whole period were minor. Even the unplanted bare treatment had a mean per-event retention of 93%, suggesting little benefit in selecting plants with high water use to improve retention performance.

The relatively high retention in our study resulted primarily from the large number of small rainfall events (the 50th percentile event rainfall depth was 2 mm). These small events (< 2 mm) produced no runoff which has been observed elsewhere (Stovin et al., 2012; Voyde et al., 2010). This has also been demonstrated in Sheffield, UK, where high event-based retention was also due to the large number of small events (Berretta et al., 2014; Poë et al., 2015; Stovin et al., 2012, 2013, 2015). For small events, rainfall is likely intercepted by the vegetation and subsequently evaporated (Dunnett et al., 2008; MacIvor and Lundholm, 2011; Nagase and Dunnett, 2012) so that the first 5 mm of rainfall was often retained as the initial absorption (Carter and Rasmussen, 2006). Therefore, system storage and *ET* are likely not important factors for small, isolated events, e.g., events < 2 mm (Fassman-Beck et al., 2013). The large influence of event size most likely resulted in the minimal differentiation in retention among planting treatments. We did, however, observe significantly different retention performances among plant treatments for medium (11–16.8 mm) and large (> 16.8 mm) events. These large events were

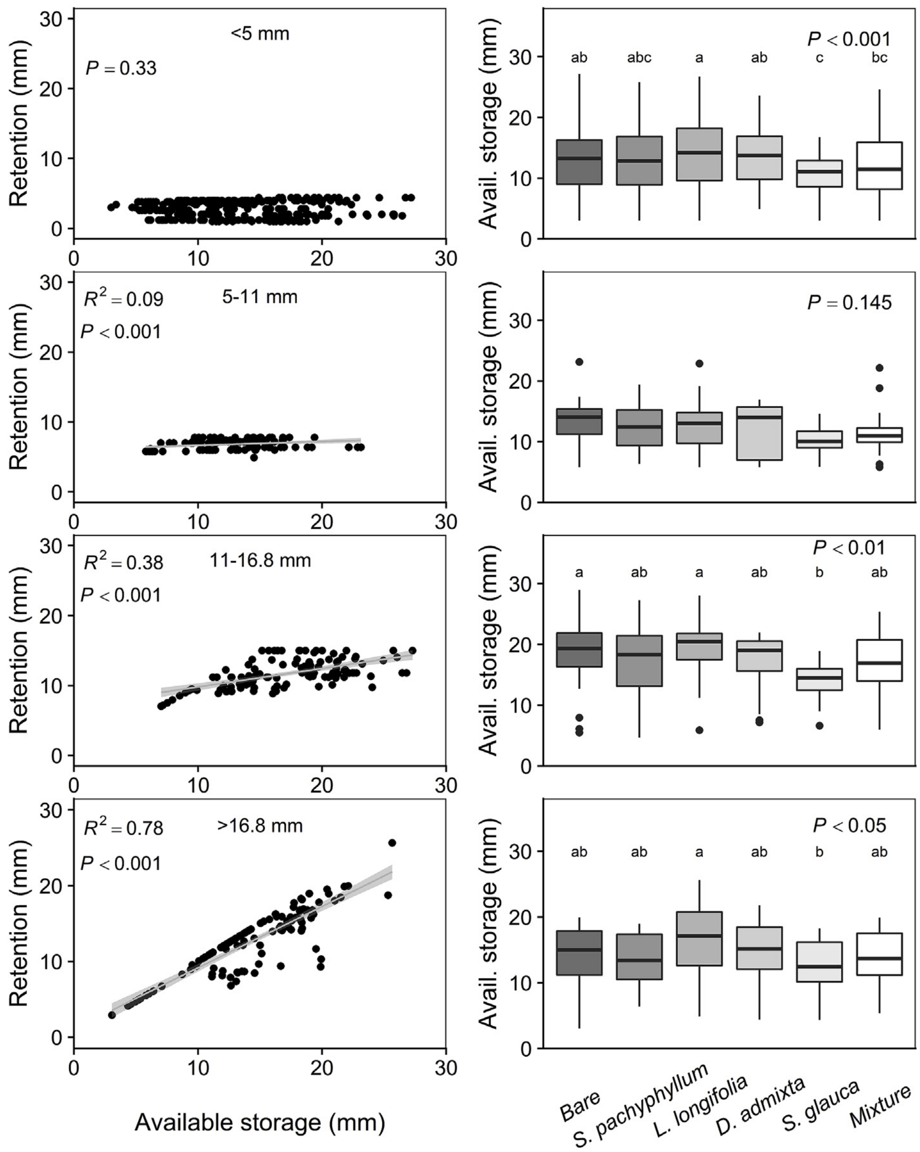

Fig. 4. For different rainfall events (< 5, 5–11, 11–16.8 and > 16.8 mm), the relationships between retention performance and available storage before events (left panel), and pre-event available storage according to plant treatment (right panel). Adjusted R^2 and *P*-values were derived from regression analysis (left panel). *P*-values and different letters indicate significant differences among plant treatment means as determined from one-way ANOVA (right panel).

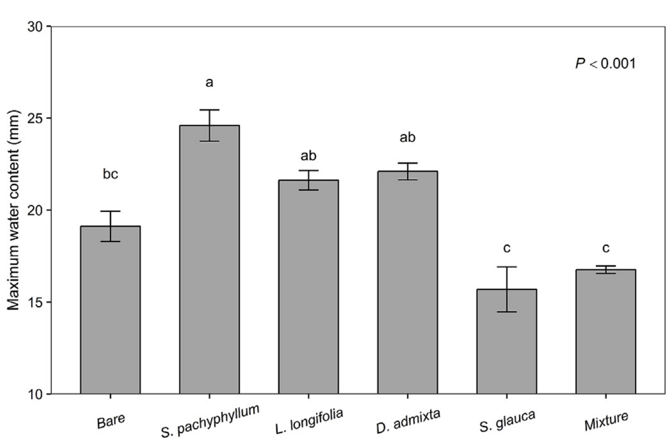

Fig. 5. Mean maximum water content for all plant treatments. Different letters denote significant differences among means (one-way ANOVA, n = 3; all $P < 0.001$) and bars represent mean standard error.

mainly in summer, when ET_o was relatively high and the antecedent dry period (ADWP) was relatively long, creating the opportunity for different plant treatments to deplete stored moisture in the substrate. This is consistent with other studies, which have also shown differences in retention efficiencies with respect to substrate or plant selection treatments to only be evident in large events (≥24.8 mm) (Berretta et al., 2014; Poë et al., 2015; Stovin et al., 2012, 2013, 2015). The influence on retention for large events from plants is particularly relevant for flooding. Therefore, even though different plant species showed similar retention performance overall due to the dominance of small rainfall events, plant selection has the potential to improve retention performance of green roofs in climates dominated by larger events, and could thus be important for flood mitigation (Lee et al., 2013).

4.2. The key plant-based drivers of green roof retention performance

Although the influence of plant treatments on retention was largely dependent on rainfall event size, plants affected retention through their influence on water storage capacity available before each rainfall event. Available storage significantly influenced retention performance except

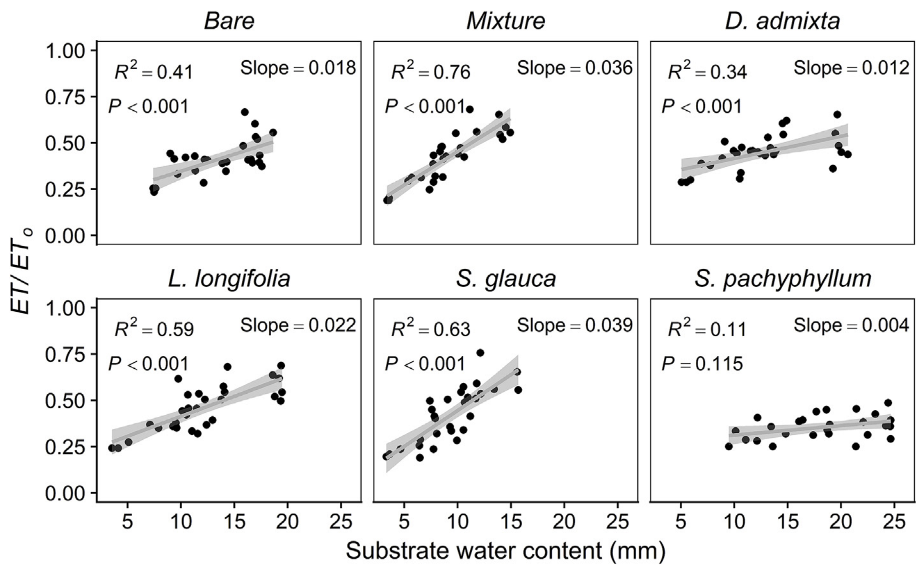

Fig. 6. Relationships between ET/ET_o and substrate water content for the six plant treatments. Lines represent the linear model and shading represents the 95% prediction interval. The adjusted R^2, *P*-value and slope were derived from regression analysis.

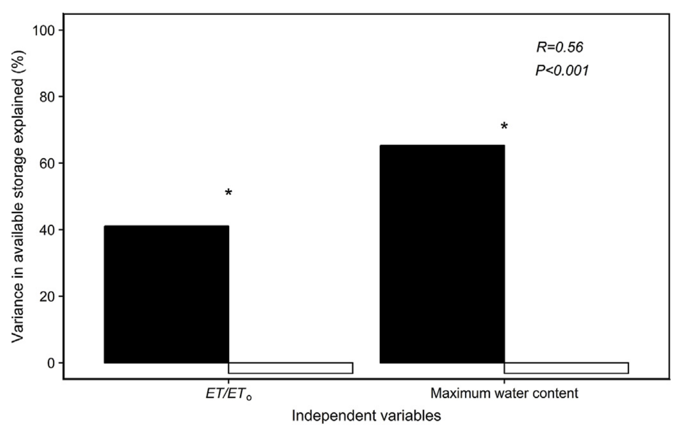

Fig. 7. Contributions of ET/ET_o and maximum water content to explanation of variance in available storage, as determined from hierarchical partitioning. R and *P*-values were derived from correlation analysis. Black and white bars show individual and joint contribution of each predictor variable to the full model, respectively. The '*' indicates the independent variables which contributed significantly to the full model ($P < 0.05$).

in the events < 5 mm. It suggested that differences among plant treatments with regard to available storage were consistent with their differences in retention performance. For example, the *S. glauca* and the mixture treatments had less available storage on average before each rainfall event and this was likely responsible for their lower retention performance. Plant species and their *ET* likely did not influence retention in small events because available storage was always greater than the rainfall depth of these events. This finding demonstrates the importance of available storage as a factor which links all parts of the moisture balance of a green roof, and therefore controls the hydrological response to rainfall events (Bengtsson et al., 2005; DeNardo et al., 2005; Sims et al., 2016; Stovin et al., 2012, 2013).

Importantly, plant treatments differed with regard to the *maximum water content*, demonstrating that not all effects of plants were *ET*-related. Differences in maximum water content among plant treatments were likely due differences in root morphology which can affect how water is stored and moves through the substrate (Carpenter and Hallam, 2009; Johnson and Lehmann, 2006; Zhang et al., 2018). In our study, *S. pachyphyllum* had the highest maximum water content, while *S. glauca* and the mixture had the lowest maximum water content. Shallow-rooted species such as *Sedum* are more likely to increase the maximum water capacity because of the binding effect of its roots in the highly porous substrate (Poë et al., 2015). However, their shallow roots limit their ability to transpire water from deeper parts of the substrate (Collins and Bras, 2007; Poë et al., 2015). In contrast, non-succulent plant species such as *S. glauca* have coarse roots which develop throughout the substrate depth (Zhang et al., 2018) in response to water and nutrient availability (Bao et al., 2014; Hodge, 2009). This root structure potentially results in a greater number of macropores and sub-surface channels that can cause uneven and rapid water movement through unsaturated parts of the substrate; i.e., preferential flow pathways (Glass et al., 1991; Hatt et al., 2009; Le Coustumer et al., 2012; Walter et al., 2000; Wardynski and Hunt, 2012; Zhang et al., 2018). Plant-induced preferential flow has been observed in these high saturated hydraulic conductivity substrates which are typically designed for high infiltration rates to avoid overflow (Gonzalez-Merchan et al., 2014; Hatt et al., 2009). Zhang et al. (2018) recently suggested preferential flow pathways were reducing the maximum storage capacity of green roof modules planted with *S. glauca*, as they observed less time to initiation of runoff. However, it is also likely that the maximum water content of these modules was decreased by roots accumulating in the drainage layer beneath the substrate. Drainage or retention layers can recharge the overlying substrate layer (Vesuviano and Stovin, 2013) and can represent a significant water storage, especially in shallow extensive green roof systems (Lu et al., 2015). Therefore, accumulation of roots in the drainage layer for modules planted with *S. glauca* may also have contributed to the observed reduction in maximum water content. Indeed, given the drainage layer in the modules had the capacity to store 8.1 mm which is equivalent to the different in maximum water content between modules planted with *S. pachyphyllum* and *S. glauca*. Therefore, in our study, while plant treatments had a significant influence on available storage, it was the result of reduced maximum storage capacity of modules, rather than differences in water use.

In our study, *S. pachyphyllum* showed a conservative water use strategy and did not adjust water use with according to substrate water availability. In contrast, *S. glauca* was far more responsive to changes in water availability. This suggests that the limitation of *S. pachyphyllum* was its water use strategy, while the limitation for *S. glauca* was the substrate not having enough water available for *ET*. The limited ET/ET_o for *S. glauca* in turn reduced the regeneration storage for the next event. In contrast, plant species such as *L. longifolia* had the ability to use more water at greater substrate moisture contents and increase available storage in the system, which resulted in both greater available storage and greater ET. This study shows that plants influence retention not only through transpiration of water that regenerates storage for the next event, but also by modifying the substrate properties. Some species, e.g., *S. glauca,* reduce the maximum storage. Based on this trade-off, the best performing plant species in this study was *L. longifolia,* which had not only had high water use at greater substrate moisture contents but was also able to maintain or increase the maximum water content of the system. The decrease in available storage in *S. glauca* and mixture modules, likely caused by preferential flow (Zhang et al., 2018), had a greater impact on available storage than did their evapotranspiration potential (ET/ET_o in relation to substrate water content). Therefore, although high water users have been suggested as the ideal plants for increasing retention on green roofs (Farrell et al., 2013b), it is also important to select plants with root systems that do not reduce the storage capacity of the substrate. We suggest that this is because storage capacity is the most important factor driving rainfall retention and therefore reducing runoff from green roofs.

4.3. Limitations and future research

Our study did not specifically quantify differences in canopy interception among treatments which can make a significant contribution to retention (Soulis et al., 2017b). The size and physical structure or 'architecture' of certain plants can effectively increase the storage capacity of a green roof (Morgan et al., 2013; Ouldboukhitine et al., 2012; Wadzuk et al., 2013). In the present study, the maximum storage capacity of modules planted with the high-water using species, *S. glauca* as well as the mixture were reduced. While we attribute this to roots effectively decreasing the storage capacity of the substrate, lower interception by modules planted with *S. glauca* modules may also have reduced retention, compared with other modules. However, our method did not allow us to quantify the relative influence of interception on retention, due to the need to wait 24 h after the simulated rainfall was applied to ensure accurate measurement of substrate water content and *ET*. Future studies are needed to determine the relative importance of canopy interception compared with other non-*ET* and *ET*-related plant effects on rainfall retention.

Our experimental method also limited our ability to quantify *ET* on the day rainfall was applied. Post-event module weights were taken 24 h after each simulated event to avoid generating runoff when lifting the modules to obtain their weight. As a result, any *ET* on the day of each rainfall event was not quantified. While the relationships built to describe the efficiency of each planted treatment to deplete stored moisture were not affected by this, it restricted our ability to quantify all *ET*. The fact that modules were only weighed before and after each simulated event also limited our ability to quantify temporal patterns in *ET* between simulated events. This restricted our ability to describe the decline in *ET* in relation to time since rainfall, which is used in green roof water balance models, where increasing time since rainfall is used as a proxy for declining *SWC* (Berretta et al., 2014). However, the functions we built here directly quantified ET/ET_o in relation to *SWC* which is preferable to using increasing time since rainfall as a proxy for declining *SWC* (Daly et al., 2012; Stovin et al., 2013). This is because 'time' is not standardised and is better represented by cumulative evaporative demand, and further, time since rainfall does not take into account how much rainfall fell and therefore how much moisture is available for evapotranspiration. However, approaches which quantify temporal aspects of *ET* are highly valuable and should be used where possible.

5. Conclusions

This study suggests that the maximum storage capacity defines available, pre-event storage, which in turn limits the water availability for plants (i.e., ET/ET_o). Therefore, although *S. glauca* and the mixture had the physiological potential to increase water use when substrate water content increased, their retention performance was limited by storage capacity, which was always less than *L. longifolia* and *S. pachyphyllum*. The plant-induced reduction in the storage capacity of the substrate had a greater effect on available storage and therefore retention, regardless of whether the plant was a high or low water user. From a stormwater management perspective, selecting green roof plants with high water use is more likely to improve retention performance; however, we need to be sure that these plants do not reduce the storage capacity of the substrate. Specifically, species such as *S. glauca* in current study with a fibrous, deep penetrating and vertical root system should not be selected to improve green roof rainfall retention, despite its higher potential to deplete stored moisture.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2018.11.070.

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